

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Using Python, PostgreSQL and Power BI

Prepared by: Tushar Chinubhai Valand

Table of Contents

1. Project Overview
2. Dataset Description
3. Project Objectives
4. Tools and Technologies
5. Data Preparation and Cleaning
6. Exploratory Data Analysis
7. Data Analysis Using SQL
8. Power BI Dashboard
9. Key Findings
10. Business Recommendations
11. Conclusion

1. Project Overview

Introduction

Customer shopping behaviour analysis helps businesses understand purchasing patterns, customer preferences, and spending habits. By analysing transactional data, organisations can identify valuable customer segments, improve marketing strategies, and increase customer satisfaction.

This project analyses a customer shopping dataset containing 3,900 transactions. Python was used for data cleaning and exploratory analysis, PostgreSQL was used for SQL-based business analysis, and Power BI was used to build an interactive dashboard for data visualisation.

2. Dataset Description

The dataset contains customer transaction records collected from an online retail store. It includes demographic information, product details, purchase amounts, payment methods, subscription status, and customer review ratings.

Dataset Summary

- Number of Records: **3,900**
- Number of Columns: **18**
- Missing Values: **37 (Review Rating)**
- Dataset Type: **Customer Shopping Transactions**

Insert Figure 1: Dataset Overview

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7

3. Project Objectives

The main objectives of this project are:

- Analyse customer shopping behaviour.
 - Clean and prepare the dataset for analysis.
 - Perform exploratory data analysis using Python.
 - Analyse business questions using SQL.
 - Develop an interactive Power BI dashboard.
 - Generate business insights and recommendations.
-

4. Tools and Technologies

Tool	Purpose
Python	Data Cleaning & Analysis
Pandas	Data Manipulation
PostgreSQL	Database & SQL Queries
pgAdmin 4	SQL Execution
Power BI	Dashboard Development
VS Code	Coding Environment

5. Data Preparation and Cleaning

Introduction

Before analysing the dataset, several preprocessing steps were performed to ensure data quality and consistency.

The dataset was imported into Python using Pandas and explored using functions such as `head()`, `info()`, and `describe()`. Missing values in the **Review Rating** column were identified and replaced using the median value. Column names were standardised into snake_case format,

redundant columns were removed, and new features such as **Age Group** were created to support further analysis.

Finally, the cleaned dataset was imported into PostgreSQL for SQL-based analysis.

Insert Figure 2: Data Cleaning Process

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                      3900 non-null   str
4   Category                            3900 non-null   str
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   str
7   Size                                3900 non-null   str
8   Color                               3900 non-null   str
9   Season                              3900 non-null   str
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   str
12  Shipping Type                      3900 non-null   str
13  Discount Applied                   3900 non-null   str
14  Promo Code Used                    3900 non-null   str
15  Previous Purchases                 3900 non-null   int64
16  Payment Method                     3900 non-null   str
17  Frequency of Purchases             3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	3
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	1737
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN

6. Exploratory Data Analysis (Python)

Introduction

Exploratory Data Analysis (EDA) was performed using Python to understand customer demographics, purchasing behaviour, and product performance.

The analysis revealed that:

- Young Adult customers represented the largest customer group.
- Clothing was the most frequently purchased product category.
- The average purchase amount was approximately £60.
- The average customer review rating was 3.75.
- Most customers were non-subscribers.

These findings provided the foundation for the SQL analysis.

Insert Figure 3: Python Visualisations

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

7. Data Analysis Using SQL

Introduction

The cleaned dataset was imported into PostgreSQL, where SQL queries were used to answer important business questions and generate meaningful insights.

Revenue by Gender

The analysis showed that male customers generated higher total revenue than female customers.

Insert Figure 4: SQL Query

```
SELECT
    gender,
    SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY gender;
```

Insert Figure 5: SQL Output

	gender text	total_revenue numeric
1	Female	75191
2	Male	157890

High-Spending Customers

Customers who spent above the average purchase amount while receiving discounts were identified. These customers represent valuable targets for loyalty programmes.

Insert Figure 6: SQL Query

```
SELECT *
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >
(
    SELECT AVG(purchase_amount)
    FROM customer
);
```

Insert Figure 7: SQL Output

	customer_id bigint	age bigint	gender text	item_purchased text	category text	purchase_amount bigint	location text	size text	color text	season text	review_rating double precision	subscription_status text	shipping_type text	discount_applied text
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes
2	3	50	Male	Jeans	Clothing	73	Massachuse...	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes
4	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes
5	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes
6	12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes
7	13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes
8	16	64	Male	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes	Store Pickup	Yes
9	20	66	Male	Pants	Clothing	90	Rhode Island	M	Green	Summer	3.3	Yes	Standard	Yes
10	22	31	Male	Pants	Clothing	62	North Caroli...	M	Charcoal	Winter	4.1	Yes	Store Pickup	Yes

Top-Rated Products

Products such as Gloves, Sandals, and Boots received the highest average customer ratings, indicating strong customer satisfaction.

Insert Figure 8: SQL Query

```
SELECT
    item_purchased,
    ROUND(AVG(review_rating::numeric),2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY avg(review_rating) DESC
LIMIT 5;
```

Insert Figure 9: SQL Output

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Subscriber vs Non-Subscriber Analysis

Subscribers and non-subscribers had similar average purchase amounts, but non-subscribers generated higher total revenue because they represented a larger customer base.

Insert Figure 10: SQL Query

```
SELECT
    subscription_status,
    COUNT(customer_id) AS total_customers,
    ROUND(AVG(purchase_amount),2) AS avg_spend,
    ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

Insert Figure 11: SQL Output

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

8. Power BI Dashboard

Introduction

An interactive Power BI dashboard was developed to present the results of the analysis in a clear and user-friendly format. The dashboard includes key performance indicators, charts, and filters that allow users to explore customer behaviour and sales trends.

The dashboard displays:

- Total Customers
- Average Purchase Amount
- Average Review Rating
- Sales by Category
- Revenue by Age Group
- Gender Distribution
- Subscription Status
- Shipping Type

Insert Figure 12: Power BI Dashboard



9. Key Findings

The analysis identified several important business insights:

- Male customers generated higher revenue than female customers.
 - Clothing was the highest-selling product category.
 - Young Adult customers were the largest customer segment.
 - The average purchase amount was approximately **£60**.
 - The average customer review rating was **3.75**.
 - Products such as Gloves, Sandals, and Boots received the highest customer ratings.
 - Non-subscribers contributed more total revenue due to their larger customer base.
-

10. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Increase subscription membership by offering exclusive benefits.
 - Promote highly rated products through targeted marketing campaigns.
 - Reward loyal customers with personalised offers and discounts.
 - Use customer purchasing behaviour to create personalised product recommendations.
 - Improve customer retention through loyalty programmes and promotional campaigns.
-

11. Conclusion

This project demonstrated a complete data analytics workflow using Python, PostgreSQL, and Power BI. The dataset was cleaned, analysed, and visualised to identify customer purchasing patterns and business opportunities.

Python was used for data preparation and exploratory analysis, PostgreSQL was used to answer business questions through SQL queries, and Power BI was used to create an interactive dashboard for effective data visualisation.

The insights generated from this project can support better decision-making, improve customer engagement, and help businesses optimise their sales and marketing strategies.